Spatial Pooling Transformer Network and Noise-Tolerant Learning for Noisy Hyperspectral Image Classification

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***Abstract*— Hyperspectral image (HSI) classification is a hot topic in remote sensing (RS). A large number of studies have been proposed and achieved excellent performance. Most of them rely on accurate annotations. However, this requirement cannot always be met. Due to the complex contents within HSIs and the uncontrollable external interference factors, incorrect labels are inevitable. Thus, the study of noisy HSI classification is boomed. Some attempts have been made, and their central ideas are to filter the noisy samples from the training set. Although feasible, this would result in information loss, i.e., the contents covered by the removed samples are ignored. Besides, the characteristics of HSIs are not fully considered in many models. To overcome the above limitations, we develop a spatial pooling transformer network (SPTNet) and a noise-tolerant learning algorithm (NTLA) in this article. SPTNet first uses a spectral feature extraction (SFE) module to capture the rich spectral information from HSI patches. Then, three spatial pooling transformers (SPTs) are constructed and stacked to explore the spatial knowledge and depress confusing clues caused by the HSI patch division. Finally, a standard transformer encoder is used to enhance the obtained spectral–spatial features for the downstream classification. To use SPTNet to handle noisy HSI classification, the noise-tolerant learning algorithm is designed. It encloses two parts, i.e., a data partition scheme and a label-independent similarity regularization. The data partition scheme divides the training data into clean and noisy sets. Then, the clean samples are used to train SPTNet with the classification loss function. At the same time, similarity regularization helps SPTNet to comprehensively understand HSIs by analyzing the resemblance between clean and noisy samples. Integrating two parts into a cotraining framework, SPTNets can be trained under a noisy scenario. Four popular HSI datasets are selected to testify to our methods. The positive results demonstrate that the combination of SPTNet and the noise-tolerant learning algorithm is helpful to the noisy HSI classification. Our source codes are available at https://github.com/TangXu-Group/Hyperspectral-Images-Classification/tree/main/SPTNet-NTLA.**

***Index Terms*— Hyperspectral image (HSI) classification, noisy labels.**

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I. Introduction

In RECENT years, with the rapid development of remote sensing (RS) technology, hyperspectral images (HSIs) have gained growing attention [1]. Owing to the hundreds of narrow and contiguous spectral bands, HSIs contain gorgeous spatial–spectral information, which is conducive to analyzing the diverse land covers [2]. Among various HSI image processing techniques, HSI classification is a fundamental one aiming to provide semantic labels for each pixel within an HSI [3]. The solid classification results can be applied to numerous RS applications, such as agricultural administration [4], ecological observation [5], and so on.

Over the last decades, many HSI classification methods have been proposed [6], [7], [8]. In the beginning, scholars completed the classification in two steps [9], i.e., low-level visual feature extraction and classifier development based on traditional machine learning. At that time, the classification results are acceptable but satisfactory. The reasons are twofold. One is the contradiction between the complex contents within HSIs and the low representation capacity of handcrafted visual features. The other one is that the independent classifiers cannot capture the feature distributions well. When deep learning appears, the HSI classification is suddenly enlightened [10], [11]. Diverse deep neural networks, such as autoencoder [12], recurrent neural networks (RNNs) [13], and convolutional neural networks (CNNs) [14], [15], are adopted and developed to deal with HSI classification tasks [16]. Owing to the strong feature learning capacity and combination of feature extraction and classification, these methods achieve cracking classification performance. Recently, visual transformer [17], [18] enters the HSI community. In contrast to CNNs and other deep models, the visual transformer is good at modeling the long-distance relations within HSIs. Thus, the global contextual information hidden in HSIs can be captured, which is good for the final classification [19], [20], [21], [22], [23], [24].

Although the above methods have succeeded in HSI classification, they almost depend on an assumption, i.e., the given manual labels are correct. In other words, the training samples

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have credible ground truths, which can guide the model to capture valuable information from HSIs for the downstream classification. Nevertheless, this hypothesis is not always true in practice [25]. Due to the complex contents of HSIs and their high-dimensional properties, incorrect annotation is inevitable [26]. Consequently, developing robust classification methods with the consideration of the negative influence of noisy labels is an urgent and necessary task. Despite some rewarding attempts that have been made on this topic [27], [28], [29], there is still much potential for improvement. First, most existing methods are habituated to adopting CNNs as the feature extraction backbones. Thus, the local clues within HSIs can be fully explored. However, the global contexts hidden in HSIs are always ignored, limiting their performance more or less. Second, selecting and removing the possible noise samples are critical in the current models. This is feasible; however, some samples with clean labels would also be eliminated, leading to information loss. Besides, the noisy sample filtering scheme may aggravate the imbalance issue in HSI classification. Last but not least, although the noisy samples have incorrect labels, the contents they covered are still valuable for understanding HSIs. Nevertheless, most models only focus on their negative influence but neglect their positive functions, leading the model cannot capture the contents completely.

To overcome the above limitations, we propose a spatial pooling transformer network (SPTNet) and a noise-tolerant learning algorithm (NTLA) in this article. The function of SPTNet is to fully explore the valuable spectral–spatial information from HSIs. To this end, besides adopting 2-D and 3-D CNNs, a spatial pooling transformer (SPT) is developed. Combining them allows the knowledge hidden in HSIs to be mined from local and global aspects. The function of the noise-tolerant learning algorithm is to ensure that the SPTNet can work smoothly under a noisy scenario. To achieve this goal, a data partition scheme is first designed to select the clean samples for training SPTNet with the classification loss function. Besides, instead of a single SPTNet, the algorithm leverages two SPTNets at the same time to enhance the creditable of selected clean samples. Then, a similarity regularization is developed to mine information from both clean and noisy samples, avoiding the information loss caused by the noisy sample filtering. After integrating the above parts, SPTNet can be trained with noisy data.

The main contributions of this article are summarized as follows.

1) A robust noisy HSI classification framework is proposed, including a feature extraction network (i.e., SPTNet) and a noise-tolerant algorithm. Accurate classification results can be achieved by combining discriminative spatial–spectral features and specific processing methods for noisy labels.
2) SPTNet is a simple model mainly involving 2-D/3-D CNNs and SPTs. CNNs take charge of capturing the spectral information from HSIs, while the SPTs focus on organizing the spatial information of HSIs and mitigating the redundant information introduced by HSI patches. This way, the valuable features of HSIs can be obtained, and the specific characteristics of HSI classification tasks can be considered.
3) A specific learning algorithm is designed to guarantee SPTNet's performance under noisy conditions. Not only the statistical model but also the advanced learning paradigms are incorporated into our algorithm. Consequently, the negative impacts of noisy samples can be whittled down, and the contents in HSIs can be interpreted accurately and thoroughly.

The remaining parts of this article are organized as follows. Section II reviews some published studies on HSI classification, noisy label learning, and HSI classification with noisy labels. In Section III, our SPTNet and the noise-tolerant learning algorithm will be introduced in detail. Experiment results on four public HSI datasets are discussed in Section IV. Finally, a short conclusion is drawn in Section V.

II. Related Work

In this section, we review the published works from the following aspects, i.e., HSI classification, learning with noisy labels, and HSI classification with noisy labels.

A. HSI Classification

Nowadays, CNNs and their variants have dominated the HSI classification community. Benefiting from the stacked structure and nonlinear fitting talents, CNNs can explore the spatial–spectral information from HSIs for accurate classification results [30], [31], [32], [33]. Considering the cube properties of HSIs, a 3-D octave convolution network with a spatial–spectral attention mechanism was proposed in [34]. The authors first expand the original octave convolution into a 3-D version to explore the spatial–spectral clues from HSIs with a relatively time cost. Then, the spatial–spectral attention mechanism is developed to enhance the obtained features mutually. The reported results counted on four HSIs confirm its usefulness. Lv et al. [35] introduced an efficient spatial–spectral interaction network to speed up the classification process. Instead of using normal convolution, a lightweight operator named involution is developed to focus on spectral messages with the consideration of redundant information reduction. Also, a dual-pooling layer is proposed to pay attention to spatial clues. Based on them, the HSI classification can be completed efficiently. Besides the supervised models mentioned above, some semisupervised and unsupervised methods were also developed. For instance, based on the graph convolution network (GCN), a semisupervised network was presented in [36] named dynamic multiscale graph convolutional network classifier (DMSGer). Similar to common GCN-based models, DMSGer leverages the superpixel technique to construct a small graph. Different from common GCN-based models, DMSGer updates pixel-level features rather than the superpixel-level representations directly. Finally, a simple label propagation scheme can achieve classification results. Zhang et al. [37] proposed an unsupervised CNN feature extraction network for HSI classification to release the burdens of manual labels. By integrating the simple noniterative clustering algorithm, convolutional

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encoder–decoder, and the random forest classifier, favorable results can be obtained.

Recently, since the transformer is good at modeling the long-distance relations hidden in images, many methods have been developed based on it for diverse RS tasks [38], [39], [40], [41]. Unlike the fixed receptive field of CNNs, the transformer uses the multihead self-attention (MHSA) to model a global receptive field. In the HSI classification field, the transformer is also prevalent. In [42], a group-aware hierarchical transformer was developed. By merging MSHA and a grouped pixel embedding (GPE) module, the spatial–spectral features of HSIs can be learned in a local–global way. Then, the accurate classification results are generated by a hierarchical combination of MSHA and GPE. Bai et al. [43] proposed a multibranch transformer for HSI classification, which involves a spatial attention module and a multibranch prediction method. The spatial attention module aims to explore the positional relationships in the HSI patches, while the multibranch prediction method pays attention to enhancing the final representations. The positive classification maps illustrate their effectiveness. Yu et al. [44] introduced a multilevel transformer network to classify HSIs. In this network, valuable spatial information is explored by transformer encoders. Also, the spectral messages are captured by a multilevel feature fusion decoder. This way, the comprehensive representation of HSIs can be obtained for the downstream classification. Zhang et al. [45] proposed a lightweight transformer network to handle HSI classification. The network includes two lightweight self-attention modules that explicitly extract local information in shallow layers and capture long-range dependencies in deep layers. Therefore, the model can reduce the amount of memory and calculation while ensuring superior classification performance. An HSI classification framework based on multiattention transformer and active learning was proposed by Zhao et al. [46], in which an outlook-attention module is combined with a self-attention module to encode fine-level features and global context relationships. Furthermore, the active learning method based on superpixel segmentation overcomes the shortage of training samples and ensures accurate classification results.

B. Learning With Noisy Labels

Learning with the noisy label first emerged in computer vision [47], [48]. Since reliable labels are difficult to obtain, this topic is significant in practice. So far, a large number of methods have been proposed, and they can be partitioned into two groups roughly. In the first group, the methods are accustomed to discovering noisy labels and decreasing their influence. For example, Shu et al. [49] developed a sample weighting algorithm to weaken the influence of noises in the training stage. The weighting function is initialized by small unbiased samples and can be updated with the model learning. The optimal weighting function and model can be obtained simultaneously by adaptively minimizing the possible noisy samples' weights. Another noisy label selection method was presented in [50]. The cotraining strategy is adopted to find the small-loss samples. Along with a specific joint loss, the noisy labels can be eliminated from the training data automatically. To further address the class imbalance caused by noisy label removal, a sample selection strategy based on contrastive learning was developed [51]. This strategy is robust to various label noises by embedding the Jensen–Shannon divergence algorithm. The noisy labels can be found and erased appropriately with the assistance of class-balance prior.

In the second group, the methods are habituated to enhancing the model's robustness to noisy labels. Based on the normal cross-entropy (CE) loss function, Wang et al. [52] designed a symmetric CE (SCE) loss to boost the robustness of the model to noisy labels. SCE consists of a CE term and a reverse CE term, which focus on achieving convergence and tolerating the noises, respectively. After applying this simple yet useful loss function, the model can be trained with noisy labels smoothly. Ma et al. [53] proposed another loss function for model training under the noisy scenario, named active passive loss (APL). APL synchronously takes the labeled and other classes into account and combines active and passive loss functions together for model training. The positive experimental results confirm the usefulness of APL. Inspired by the nearest-neighbor theory, a new learning algorithm was developed in [54]. Considering the similarities between samples, the specific learning algorithm can deal with the possible noises distinctly.

C. HSI Classification With Noisy Labels

Scholars in the HSI classification community also pay attention to the noisy label learning topic. As early as 2018, Jiang et al. [27] proposed a random label propagation algorithm (RLPA) for HSI classification with noisy labels. Based on random label propagation and ensemble learning, the noisy samples can be selected, and their labels can be corrected for the final classification. With the help of the density peak clustering algorithm, a noisy label detection method was developed in [55]. The algorithm first calculates the similarities between training samples. Then, a local density-based decision function is applied to search the potential noises. Similar work was also published in [56], where the energy of training data is analyzed for selecting the noises. Another label noise-cleaning method was presented in [57]. HSIs are first oversegmented into superpixels for constructing a graph. Then, the noise elimination problem is transformed into a label propagation task and accomplished by a graph constraint. The reported results on several HSIs illustrate that noisy labels can be removed justly. Xu et al. [58] developed a superpixel-guided sample selection network in which the noise selection and correction are integrated. On the one hand, the samples with high loss values are regarded as possible noisy samples. On the other hand, superpixels are used to correct their labels. In this way, the information loss caused by the elimination can be avoided, and classification accuracy can be guaranteed. Instead of finding and abolishing the noisy labels, a dual-channel residual network (DCRN) was proposed in [59]. Owing to the residual and dual-branch structure, the network can explore the relations among interclass/intraclass samples. Besides, a specific loss function is utilized to reduce

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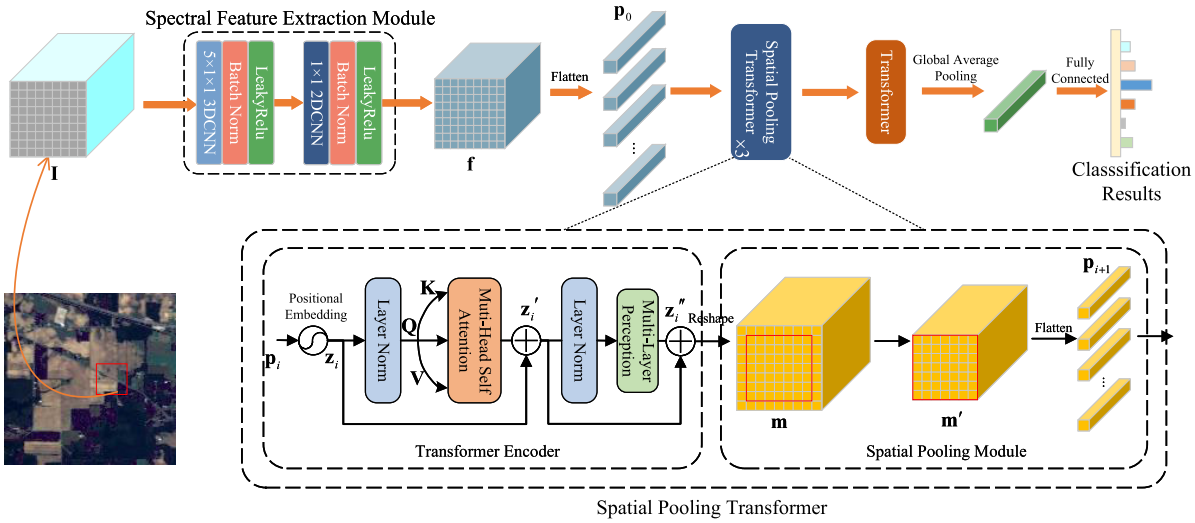


Fig. 1. Proposed SPTNet. The details of the SPT are shown in the dotted box on the bottom right.

the influence of noisy labels as much as possible. By combining them together, remarkable HSI classification results can be ensured. A triple contrast representation learning (TCRL) method was proposed by Zhang et al. [60] for noisy HSI classification. Based on different enhancement transformations, corresponding learning losses are used to explore cluster-level, instance-level, and structure-level representations, effectively reducing the negative impact of noisy samples and ensuring accurate classification in noisy scenes.

III. Method

A. Problem Statement

Before introducing the proposed SPTNet and the noise-tolerant learning method, the problem that this article needs to solve is defined in this section. For an HSI with c classes, suppose there is a set of pixels $\mathbf{X} = \{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n\}$ with semantic labels $\mathbf{Y} = \{\mathbf{y}_1, \mathbf{y}_2, \ldots, \mathbf{y}_n\}$, where $\mathbf{x}_i$ indicates the b-dimensional spectral representation of the ith pixel and $\mathbf{y}_i$ denotes the one-hot label vector with a dimension of c. Generally speaking, we believe in the correctness of labels and use these labeled pixels to train the classification model directly under the supervised learning paradigm. In the current deep era, the popular classification models are deep neural networks (e.g., CNNs and transformers), and the principal loss function is the CE loss. However, the problem we are facing is that some labels are incorrect. In other words, some pixels are misannotated manually. To depict this situation, we define another set of labels $\bar{Y} = \{\bar{y}_1, \bar{y}_2, \ldots, \bar{y}_n\}$. When $\bar{y}_i \neq \mathbf{y}_i$, $\bar{y}_i$ can be regarded as a noisy label. Also, the ratio of noisy labels' numbers to whole labels' volume is defined as the noisy rate r. In this case, the common deep networks with a CE loss would be out of work. This is because they easily overfit the noisy samples [48]. Therefore, specific deep models or subtle learning strategies should be developed to handle HSI classification with noisy labels.

B. Spatial Pooling Transformer Network

Considering the characteristics of HSIs, we construct SPTNet in this section, whose architecture is shown in Fig. 1. First, SPTNet leverages a spectral feature extraction (SFE) module to capture spectral information from HSIs. Then, three SPTs are used to explore spatial knowledge from a global aspect. Finally, a transformer encoder, a global average pooling (GAP), and a fully connected (FC) layer are adopted in serial to produce the final classification maps. Now, we introduce each component of SPTNet in detail.

Instead of an HSI, we extract patches centered on labeled pixels from the original HSI as the input of our SPTNet. The HSI patch can be denoted as $\mathbf{I} \in \mathbb{R}^{s\times s\times b}$, where $s \times s$ denotes the patch size and b indicates the number of spectral bands. After feeding the patch into SPTNet, it would first be mapped into an spectral feature map $\mathbf{f} \in \mathbb{R}^{s\times s\times c}$ by the SFE module. The structure of SFE can be found in Fig. 1, which consists of a 3-D CNN block and a 2-D CNN block. Each block involves a convolutional layer, a BatchNorm (BN) layer, and a leaky rectified linear unit (ReLU) layer. The kernel size for the convolutional layer in the 3-D CNN block is $5 \times 1 \times 1$. Also, the kernel size of the convolutional layer in the 2-D CNN block is 1×1. The reason for choosing such convolution kernels is that they are able to maintain the relative positions of pixels, which is beneficial to subsequent modules to mine spatial contents. Due to the structure of 3-D CNNs, when using them to process the spectral bands of HSIs, the bands are partitioned into segments first and then subjected to separate feature extraction. Relatively, 2-D CNNs leverage complete spectral bands to capture the spectral knowledge. Therefore, stacking 3-D and 2-D CNNs is beneficial to extract valuable spectral knowledge simultaneously. Here, the 3-D and 2-D convolutions numbers are 8 and 64, respectively. Thus, the output feature map $\mathbf{f}$ has 64 channels, i.e., $c = 64$.

Next, the obtained $\mathbf{f}$ will be fed into three SPTs to capture the global spatial information. Since the three SPTs are the same, we only explain the first SPT for clarity. Before inputting $\mathbf{f}$ into the first SPT, we flatten it into a sequence data $\mathbf{p}_0 \in \mathbb{R}^{N_0\times c}$, where $N_0 = s \times s$. The structure of SPT can be found at the bottom of Fig. 1. SPT encompasses a standard transformer encoder and a spatial pooling module. The transformer encoder involves a positional embedding

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operation, two layer normalizations (LNs), an MHSA, and a multilayer perception (MLP). Besides positional embedding operation, other parts are connected in a residual way. When $\mathbf{p}_0$ is input, the positional embedding operation would add a learnable vector $\mathbf{P}_e \in \mathbb{R}^{N_0 \times c}$ to it to record the positional relations. This step can be formulated as follows:

$$\mathbf{z}_0 = \mathbf{p}_0 + \mathbf{P}_e \tag{1}$$

where $\mathbf{z}_0 \in \mathbb{R}^{N_0 \times c}$. Then, the first LN and MHSA would be applied on $\mathbf{z}_0$ to generate $\mathbf{z}_0' \in \mathbb{R}^{N_0 \times c}$ that is rich in the global relationships between different elements. This step can be defined as follows:

$$\begin{aligned}
\mathbf{z}_0' &= \mathrm{MHSA}\left(\mathbf{z}_0^L\right) + \mathbf{z}_0, \mathbf{z}_0^L = \mathrm{LN}(\mathbf{z}_0) \\
\mathrm{MHSA}\left(\mathbf{z}_0^L\right) &= \mathrm{Concat}(\mathbf{head}_1, \mathbf{head}_2, \ldots, \mathbf{head}_h)\mathbf{W} \\
\mathbf{head}_i &= \mathrm{Attention}(\mathbf{Q}_i, \mathbf{K}_i, \mathbf{V}_i) = \mathrm{Softmax}\left(\frac{\mathbf{Q}_i \mathbf{K}_i^T}{\sqrt{d}}\right)\mathbf{V}_i \\
\mathbf{Q}_i &= \mathbf{z}_0^L \mathbf{W}_q^i, \mathbf{K} = \mathbf{z}_0^L \mathbf{W}_k^i, \mathbf{V} = \mathbf{z}_0^L \mathbf{W}_v^i
\end{aligned} \tag{2}$$

where $\mathbf{z}_0^L$ implies the normalized version of $\mathbf{z}_0$; $\mathbf{head}_i$ is the ith self-attention head; $\mathbf{Q}_i$, $\mathbf{K}_i$, and $\mathbf{V}_i$ are the ith key, query, and value of $\mathbf{z}_0^L$; $\mathbf{W} \in \mathbb{R}^{c \times c}$, $\mathbf{W}_q^i \in \mathbb{R}^{c \times c/h}$, $\mathbf{W}_k^i \in \mathbb{R}^{c \times c/h}$, and $\mathbf{W}_v^i \in \mathbb{R}^{c \times c/h}$ indicate learnable parameter matrices; and LN$(\cdot)$, MHSA$(\cdot)$, Concat$(\cdot)$, Attention$(\cdot)$, and Softmax$(\cdot)$ indicate the LN, MHSA, concatenation, self-attention, and softmax functions. Next, $\mathbf{z}_0'$ would be mapped into $\mathbf{z}_0'' \in \mathbb{R}^{N_0 \times c}$ by the second LN layer and MLP. MLP contains two FC layers with the Gaussian error linear unit (GELU) activation function. This process is expressed as follows:

$$\mathbf{z}_0'' = \mathrm{MLP}\left(\mathrm{LN}\left(\mathbf{z}_0'\right)\right) + \mathbf{z}_0' \tag{3}$$

where MLP$(\cdot)$ denotes the MLP function.

The current $\mathbf{z}_0''$ incorporates rich spectral information and gorgeous global spatial messages. However, some possible redundant knowledge still exists in $\mathbf{z}_0''$. This is because the pixels within an HSI patch may belong to various semantic categories. To mitigate this problem and remove the redundant knowledge as much as possible, we develop a simple yet useful spatial pooling module, whose flowchart can be found in Fig. 1. In particular, $\mathbf{z}_0''$ is first reshaped into a 2-D version $\mathbf{m} \in \mathbb{R}^{s \times s \times c}$. Based on the assumption that the closer a pixel is to the center point in terms of pixels, the higher the probability that its attributes are similar to those of the center point [61], [62], the possible redundant knowledge should be concentrated in the edge areas of $\mathbf{m}$. Thus, we retain a square area closer to its center point while deleting the peripheral areas for producing $\mathbf{m}' \in \mathbb{R}^{(s-l) \times (s-l) \times c}$. In this fashion, redundant information can be eliminated while helpful information can be retained and emphasized. Then, $\mathbf{m}'$ would be flattened into the sequence data $\mathbf{p}_1 \in \mathbb{R}^{N_1 \times c}$, where $N_1 = (s-l) \times (s-l)$. Note that to avoid unadvisedly filtering too much useful information along with the removal of redundant clues, we only remove the outermost information of the feature map. To this end, we set $l = 2$. The spatial pooling discussed above can be formulated as follows:

$$\mathbf{p}_1 = \mathrm{Flatten}\left(\mathbf{m}'\right), \mathbf{m}' = \mathrm{SPooling}(\mathbf{m}) \tag{4}$$

where Flatten$(\cdot)$ and SPooling$(\cdot)$ indicate the flatten and spatial pooling operations.

After three SPTs, the valuable features can be fully explored from HSI patches. Then, a transformer encoder is embedded at the top of SPTs to retain more effective information, thereby enhancing the diversity of feature descriptions and obtaining a more generalized representation. Along with a GAP layer and an FC layer, the final classification maps can be produced.

Overall, SPTNet is a simple model. Its core component (i.e., SPT) is also a straightforward module that merely combines a standard transformer encoder and a spatial pooling module. However, they fully consider the characteristics of HSIs and the specifics of HSI land-cover classification. First, SPTNet can explore the spectral–spatial information within HSIs, which is important to the classification. By applying the 2-D and 3-D convolutions to HSIs, SPTNet can capture the spectral features of HSIs from the local and global aspects. Meanwhile, SPTNet leverages three SPTs and a normal transformer encoder to analyze the spatial knowledge within HSIs from the global perspective. Second, since the input data of SPTNet are HSI patches cut from the original HSI, the pixels within a patch may belong to various semantics. The pixels that differ from the central pixel in terms of their semantics create redundant information that interferes with the model's ability to extract valuable information from HSIs. To mitigate this problem and remove the redundant knowledge as much as possible, a simple yet useful spatial pooling module is developed. Combining it with the standard transformer encoder (i.e., SPT), the spatial clues can be discovered, and the interference messages can be filtered. Although SPTNet is simple, its performance is competitive in both noisy and noise-free conditions. The details can be found in Sections IV-C and IV-D.

C. Noise-Tolerant Learning Algorithm

Although SPTNet can classify HSIs normally, its performance drops sharply when noisy samples exist. To guarantee SPTNet's behavior under the noisy scenario, we develop a noise-tolerant learning algorithm in this section, whose flowchart for a training epoch is shown in Fig. 2.

An intuitive approach for whittling down the negative influence of noisy samples is to select and remove them from the training set. According to the conclusions in [58] and [63], the loss values of clean samples obtained by deep models are always lower than those of noisy samples in the early model training stage. Therefore, we design a simple data partition scheme with the assistance of a two-component Gaussian mixture model (GMM) [64] to distinguish the possible noisy samples hidden in the training set. Here, GMM is chosen because of its strong ability in the sharpness of distribution [65]. Thus, compared with other statistical models, GMM excels at discriminating between noise and clean samples.

In particular, we first feed all training samples $(\mathbf{X}, \bar{Y})$ into SPTNet and get their loss values. Then, the two-component GMM is used to fit these losses, and the Gaussian component with a smaller mean value is deemed as the distribution of clean samples. After calculating the posterior probability of

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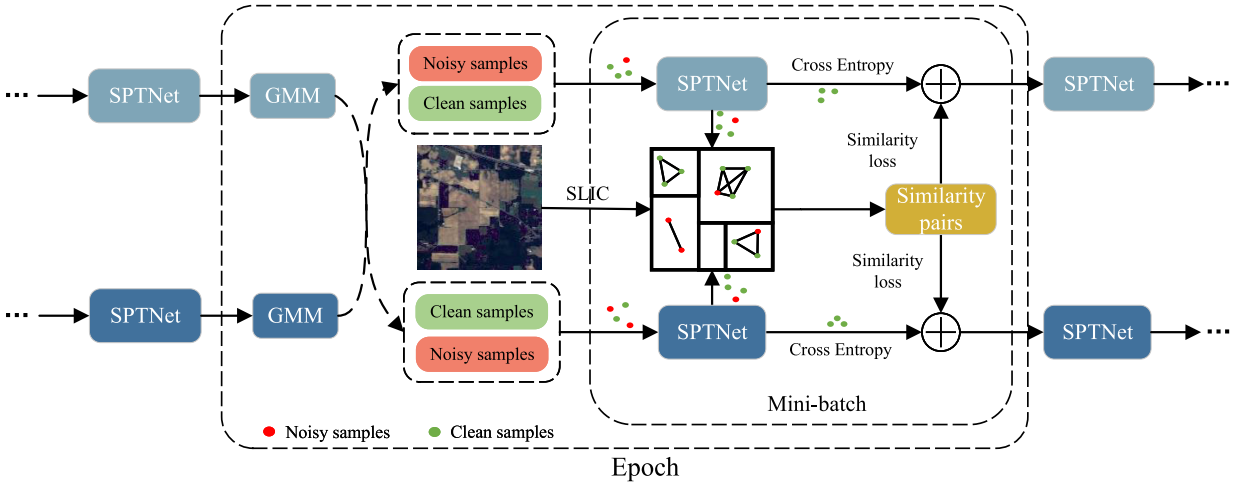


Fig. 2. Schematic of each training session of the proposed SRSPTNet. The upper and lower branches represent the process of cotraining. The similarity loss is obtained in the part between the two branches. The two jointly participate in the update of the network.

each sample belonging to the small-mean Gaussian component w_i, we can divide $\mathbf{X}$ into a clean set $\mathbf{X}^c$ and a noisy set $\mathbf{X}^n$ using a threshold τ. This process can be defined as follows:

$$\begin{cases} (\mathbf{x}_i, \bar{y}_i) \in \mathbf{X}^c, & w_i \geq \tau \\ (\mathbf{x}_i, \bar{y}_i) \in \mathbf{X}^n, & w_i < \tau. \end{cases} \tag{5}$$

Next, the samples in the clean set $\mathbf{X}^c$ would be adopted to update the SPTNet's parameters in a mini-batch way for the final classification. Note that the data partition is carried out only once per epoch, and our SPTNet will be warmed up before the first partition. Here, the warm-up denotes that we use all training data to train SPTNet along with the normal CE loss function.

Although the above method can select clean samples, noisy data will inevitably be mixed. With the increase of training rounds, the self-selection of a single SPTNet will gradually accumulate errors and cause performance degradation. To mitigate this problem, we introduce the cotraining mechanism [66] here. In detail, two SPTNets are used to select two sets of clean samples $\mathbf{X}_1^c$ and $\mathbf{X}_2^c$, independently. Then, the clean samples in $\mathbf{X}_1^c/\mathbf{X}_2^c$ are adopted to update the second/first SPTNet's parameters mutually by counting the CE losses, whose definitions are

$$\begin{aligned} \mathcal{L}_{\mathrm{CE}}^1 &= -\frac{1}{|\mathbf{X}_2^c|} \sum_{\left(\mathbf{x}_i^2, \bar{y}_i^2\right) \in \mathbf{X}_2^c} \bar{y}_i^2 \cdot \log\left(f\left(\mathbf{x}_i^2, \theta_1\right)\right) \\ \mathcal{L}_{\mathrm{CE}}^2 &= -\frac{1}{|\mathbf{X}_1^c|} \sum_{\left(\mathbf{x}_i^1, \bar{y}_i^1\right) \in \mathbf{X}_1^c} \bar{y}_i^1 \cdot \log\left(f\left(\mathbf{x}_i^1, \theta_2\right)\right) \end{aligned} \tag{6}$$

where $(\mathbf{x}_i^1, \mathbf{y}_i^1)$ and $(\mathbf{x}_i^2, \mathbf{y}_i^2)$ imply the clean samples within $\mathbf{X}_1^c$ and $\mathbf{X}_2^c$, $f(\cdot)$ means the mapping function of SPTNet, θ_1 and θ_2 are the parameters of the first and second SPTNets, and $|\mathbf{X}_1^c|$ and $|\mathbf{X}_2^c|$ denote the sizes of two clean sets.

So far, we have reduced the negative influence of the samples with incorrect labels for training SPTNet. Nevertheless, there is another point we should pay attention to, that is, the information covered by those noisy samples is missing, which prevents SPTNet from fully understanding HSIs. Especially, when the noise rate is high, a large number of samples would be filtered out, which is adverse to capturing rich knowledge from HSIs comprehensively. To overcome this limitation, we develop a similarity regularization inspired by contrastive learning [67], [68]. It aims to help SPTNet explore the messages contained in all samples without considering the labels. The basic idea of our similarity regularization is that similar pixels should have similar predictions. In other words, the feature representations of similar pixels should be more consistent. To this end, we first construct a similar pixel pair set and then design the proper loss function for these pairs.

Specifically, the simple linear iterative clustering (SLIC) [69] algorithm is applied to segment an HSI into M homogeneous regions $\{\xi_1, \xi_2, \ldots, \xi_M\}$. Owing to the characteristics of the superpixel, the pixels within a superpixel ξ_m share comparable attributes. In addition, those pixels are adjacent in spatial. Thus, they can be regarded as similar pixels. According to the above discussion, we can construct two similar pixel pair sets $\mathcal{H}_1$ and $\mathcal{H}_2$ for two SPTNets, whose definitions are

$$\begin{aligned} \mathcal{H}_1 &= \left\{ \left(\mathbf{p}_i^1, \mathbf{p}_j^1\right) \middle| \left(\mathbf{x}_i, \mathbf{x}_j\right) \in \xi_m \right\} \\ \mathcal{H}_2 &= \left\{ \left(\mathbf{p}_i^2, \mathbf{p}_j^2\right) \middle| \left(\mathbf{x}_i, \mathbf{x}_j\right) \in \xi_m \right\} \end{aligned} \tag{7}$$

where $i \neq j$, and $(\mathbf{p}_i^1, \mathbf{p}_j^1)$ and $(\mathbf{p}_i^2, \mathbf{p}_j^2)$ represent the probability vectors of $(\mathbf{x}_i, \mathbf{x}_j)$ generated by the first and second SPTNets. At the same time, considering the mutual interaction of two SPTNets, another similar pixel pair set $\mathcal{H}_3$ can also be constructed, and it can be formulated as follows:

$$\mathcal{H}_3 = \left\{ \left(\mathbf{p}_i^1, \mathbf{p}_j^2\right) \middle| \left(\mathbf{x}_i, \mathbf{x}_j\right) \in \xi_m \right\} \tag{8}$$

where i can be equal to j. The final similar pixel pair set is denoted as $\mathcal{H} = \mathcal{H}_1 \cup \mathcal{H}_2 \cup \mathcal{H}_3$. Note that the SLIC method only needs to be executed once in our learning algorithm.

After obtaining the similar pixel pairs, we develop a similarity loss to mine the profitable clues from them. Specifically, given the training data$\{(\mathbf{x}_i, \bar{y}_i)\}_{i=1}^n$, we can construct a similarity label matrix $\mathbf{A} \in \mathbb{R}^{2n \times 2n}$, whose definition is

$$\begin{cases} A_{uv} = 1, & (\mathbf{p}_u, \mathbf{p}_v) \in \mathcal{H} \\ A_{uv} = 0, & \text{otherwise}. \end{cases} \tag{9}$$

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This matrix indicates whether pairs of pixels are similar or not. Following [68], we measure the similarity of pixel pairs by computing the inner product of the probability vectors of two pixels, i.e., $S_{uv} = \mathbf{p}_u^T \cdot \mathbf{p}_v$, which can be regarded as the predicted similarity posterior probability. Thus, the similarity prediction issue can be transformed into a binary classification problem. Therefore, the similarity loss is expressed as follows:

$$\mathcal{L}_s = -\frac{1}{(2n)^2} \sum_{u=1}^{2n} \sum_{u \neq v} A_{uv} \log S_{uv} + (1 - A_{uv}) \log(1 - S_{uv}). \tag{10}$$

The similarity loss ensures that pixels from homogenous regions have similar probability distributions and enhances the information interaction between the two networks. In this way, the knowledge covered by the noisy samples can be mined by our SPTNet. Moreover, since this loss is label-independent, it is robust to label noise inherently.

Based on the above discussion, the final loss functions for two SPTNets are

$$\begin{aligned} \mathcal{L}_1 &= \lambda_s \mathcal{L}_s + (1 - \lambda_s) \mathcal{L}_{\mathrm{CE}}^1 \\ \mathcal{L}_2 &= \lambda_s \mathcal{L}_s + (1 - \lambda_s) \mathcal{L}_{\mathrm{CE}}^2 \end{aligned} \tag{11}$$

where λ_s is a free parameter to balance the importance of the CE loss and similarity loss. In the test phase, we average the predictions of the two models as the final prediction to obtain more refined results.

The noise-tolerant learning algorithm discussed in this section is summarized in Algorithm 1.

Algorithm 1 Noise-Tolerant Learning Algorithm

Input: Training data $\mathbf{X}$ and noisy label $\bar{\mathbf{Y}}$, two SPTNets θ_1 and θ_2, training rounds E_{all}, warm-up rounds E_w.

```
for epoch=1 to E_all do
  if epoch ≤ E_w then
    θ1 ← warmup(X, Ȳ, θ1)
    θ2 ← warmup(X, Ȳ, θ2)
  else
    X1^c, X1^n ← GMM(X, Ȳ, θ1)
    X2^c, X2^n ← GMM(X, Ȳ, θ2)
    Divide mini-batch X_d from X, d ∈ {1, ..., D}
    for d=1 to D do
      Calculate the CE loss L_CE^1, L_CE^2 of clean samples
      in the current batch by Eq. 6.
      Calculate the similarity loss L_s of the current batch
      by Eq. 10.
      Obtain the final losses L_1 and L_2 by Eq. 11.
      Use L_1 and L_2 to update the parameters of the two
      SPTNets respectively.
    end for
  end if
end for
```

Output: Two trained SPTNets with θ_1 and θ_2

IV. Experiments

In this section, we select four popular HSI datasets to systematically analyze the effectiveness of the proposed SPTNet and noise-tolerant learning algorithm. For convenience, we record our network and learning method as SPTNet-NTLA in the subsequent description.

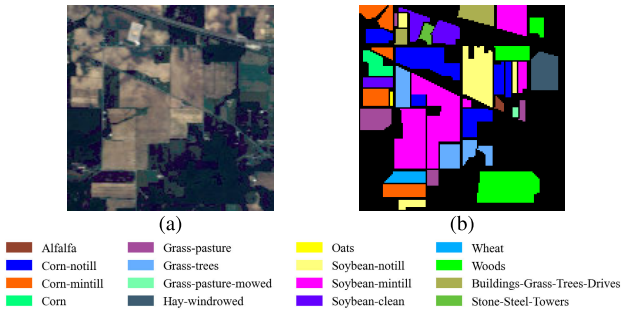


Fig. 3. Visualization of the Indian Pines dataset. (a) False-color composite image. (b) Ground-truth map. The legend below images indicates which category each color represents.

A. Datasets Description

The first one is the *Indian Pines* dataset, which was acquired by the airborne visible/infrared imaging spectrometer (AVIRIS) sensor over the Indian Pines test site in Northwest Indiana. It consists of 145×145 pixels and 220 spectral bands covering the wavelength range of 400–2500 nm. When we use this dataset to study our model and training algorithm, its spectral bands would be reduced to 200 to remove the impacts of water-absorbing regions. There are 10 249 labeled pixels, and they are divided into 16 semantic classes. The division of training and testing sets can be found in Table I, which is according to [58]. In addition, the false-color image and ground-truth map of the Indian Pines dataset are shown in Fig. 3. The second one is the *University of Pavia* dataset, which was acquired by the reflective optics system imaging spectrometer (ROSIS) sensor over Pavia in northern Italy. It has 610×340 pixels and 103 spectral bands with a spatial resolution of 1.3 m. Researchers annotate 42 776 pixels with nine semantic classes. In the following experiments, following [58], we randomly use 3930 pixels as the training data, and the rest of the labeled pixels are regarded as the testing data. The exhaustive description can be found in Table II. Also, Fig. 4 exhibits the false-color and ground-truth images. The third one is the *Houston* dataset, which was acquired by the ITRES CASI 1500 sensor in June 2012 over the campus of the University of Houston and adjacent urban areas in the United States. It encloses 349×1905 pixels and 144 reflection bands with a wavelength range of 380–1050 nm. The 15 029 pixels are labeled and partitioned into 15 land covers. As displayed in Table III, 150 categorized pixels per class are selected as the training data, and the rest are used as the testing data. This partition remains consistent with [58]. Fig. 5 shows the false-color image and ground-truth map. The last one is the *Houston 2018* dataset. It was captured by ITRES CASI 1500 at a ground sampling distance of 1 m and published in the 2018 IEEE Geoscience and Remote Sensing Society (GRSS) Data Fusion Contest. It contains 601×2384 pixels of 48 spectral bands. Among them, 504 856 pixels are labeled with 20 different semantics. As displayed

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TABLE I

NUMBER OF SAMPLES IN THE TRAINING SET AND TEST SET FOR EACH CLASS IN INDIAN PINES

No.	Class Name	Train	Test	Total
1	Alfalfa	35	11	46
2	Corn-notill	150	1278	1428
3	Corn-mintill	150	680	830
4	Corn	100	137	237
5	Grass-pasture	150	333	483
6	Grass-trees	150	580	730
7	Grass-pasture-mowed	20	8	28
8	Hay-windrowed	150	328	478
9	Oats	15	5	20
10	Soybean-notill	150	822	972
11	Soybean-mintill	150	2305	2455
12	Soybean-clean	150	443	593
13	Wheat	150	55	205
14	Woods	150	1115	1265
15	Building-grass-trees-drives	50	336	386
16	Stone-steel-towers	50	43	93
	Total	1770	8479	10249

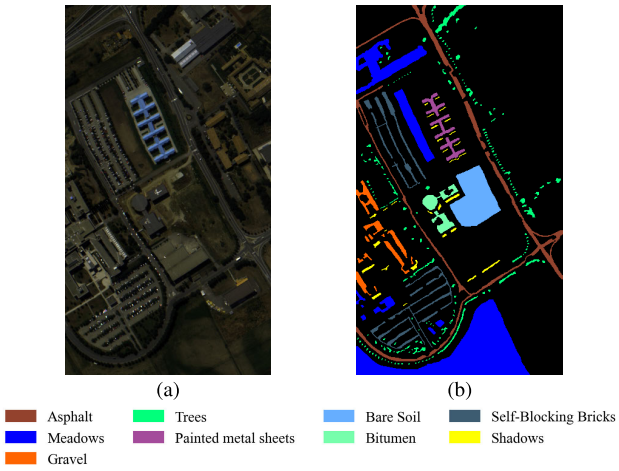


Fig. 4. Visualization of the University of Pavia dataset. (a) False-color composite image. (b) Ground-truth map. The legend below images indicates which category each color represents.

TABLE II

NUMBER OF SAMPLES IN THE TRAINING SET AND TEST SET FOR EACH CLASS IN UNIVERSITY OF PAVIA

No.	Class Name	Train	Test	Total
1	Asphalt	548	6083	6631
2	Meadows	540	18109	18649
3	Gravel	392	1707	2099
4	Trees	542	2522	3064
5	Painted metal sheets	256	1091	1345
6	Bare soil	532	4497	5029
7	Bitumen	375	955	1330
8	Self-blocking bricks	514	3168	3682
9	Shadows	231	716	947
	Total	3930	38846	42776

in Table IV, consistent with [70], 100 pixels of each class are selected as training data, and the rest are used for testing. Fig. 6 exhibits the false-color and ground-truth images. To simulate the noisy scenarios, we uniformly change labels from one class to other classes with a noise rate r [56], [57], [59].

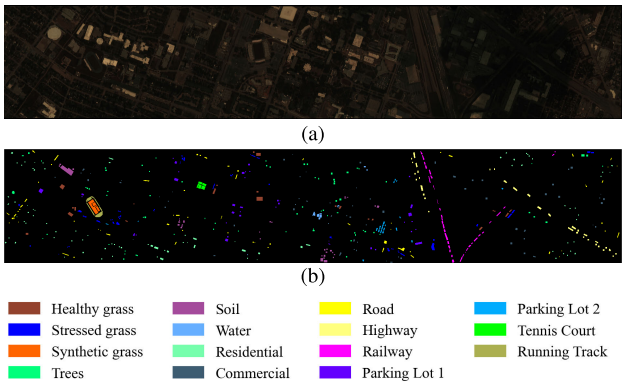


Fig. 5. Visualization of the University of Houston dataset. (a) False-color composite image. (b) Ground-truth map. The legend below images indicates which category each color represents.

TABLE III

NUMBER OF SAMPLES IN THE TRAINING SET AND TEST SET FOR EACH CLASS IN HOUSTON

No.	Class Name	Train	Test	Total
1	Healthy grass	150	1101	1251
2	Stressed grass	150	1104	1254
3	Synthetic grass	150	547	697
4	Trees	150	1094	1244
5	Soil	150	1092	1242
6	Water	150	175	325
7	Residential	150	1118	1268
8	Commercial	150	1094	1244
9	Road	150	1102	1252
10	Highway	150	1077	1227
11	Railway	150	1085	1235
12	Parking Lot 1	150	1083	1233
13	Parking Lot 2	150	319	469
14	Tennis Court	150	278	428
15	Running Track	150	510	660
	Total	2250	12779	15029

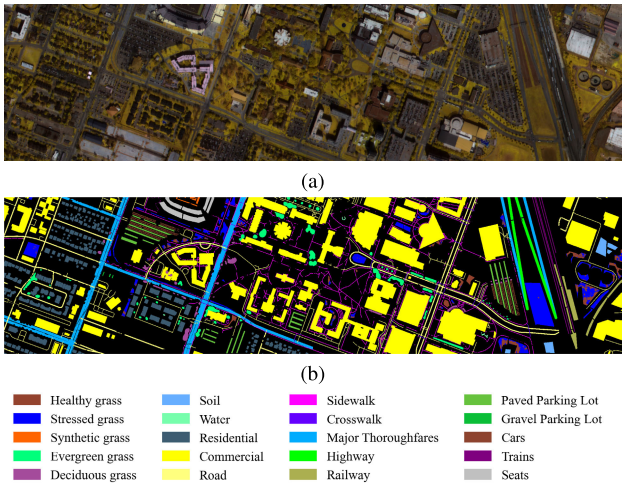


Fig. 6. Visualization of the University of Houston 2018 dataset. (a) False–color composite image. (b) Ground-truth map. The legend below images indicates which category each color represents.

B. Experiment Settings

All experiments are conducted on a workstation with a single GeForce RTX 2080 Ti of 11G memory with CUDA

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TABLE IV
NUMBER OF SAMPLES IN THE TRAINING SET AND TEST SET FOR EACH CLASS IN HOUSTON 2018

No.	Class Name	Train	Test	Total
1	Healthy grass	100	9699	9799
2	Stressed grass	100	32402	32502
3	Synthetic grass	100	584	684
4	Evergreen grass	100	13495	13595
5	Deciduous grass	100	4921	5021
6	Soil	100	4416	4516
7	Water	100	166	266
8	Residential	100	39672	39772
9	Commercial	100	223652	223752
10	Road	100	45766	45866
11	Sidewalk	100	33929	34029
12	Crosswalk	100	1418	1518
13	Major Thoroughfares	100	46248	46348
14	Highway	100	9765	9865
15	Railway	100	6837	6937
16	Paved Parking Lot	100	11400	11500
17	Gravel Parking Lot	100	46	146
18	Cars	100	6447	6547
19	Trains	100	5269	5369
20	Seats	100	6724	6824
	Total	2000	502856	504856

11.0 and Intel Xeon Silver 4210 CPU. The compilation platform used in this article is PyTorch. The parameters of SPTNet are initialized randomly. When we apply the noise-tolerant learning algorithm to train SPTNets, the Adam optimizer is selected to update SPTNets' parameters. The training epochs and the initial learning rate are 200 and $2e-3$, and the learning rate decreases by $0.8\times$ every ten epochs. The warm-up cycle equals 20/20/30/30 for the Indian Pines, University of Pavia, Houston, and Houston 2018 datasets. To study the effectiveness of the proposed noise-tolerant algorithm, we first select five popular HSI classification networks, including 3-D CNN [71], spectral–spatial residual network (SSRN) [72], SpectralFormer [20], local transformer with spatial partition restore (SPRLT) [24], and spectral–spatial feature tokenization transformer (SSFTT) [21]. Then, the usual learning manner and our noise-tolerant learning algorithm are used to train them separately under the noisy scenario. Their classification results counted on the noisy Houston 2018 dataset are exhibited in the below table. Our SPTNet's results are also displayed for reference. Note that the free parameters of the selected networks are set according to the original literature, and the experimental settings maintain consistency with the contents mentioned in Section IV-B. Some free parameters should be set in advance. For SPTNet, the free parameters are the patch size $s \times s$, the volume of attention heads h in the transformer encoder, the dimension of the features c generated by SFE, and the number of SPT n_s. For the noise-tolerant learning algorithm, the free parameters include the hyperparameter λ_s for the final loss (11), the threshold τ in the data partition scheme (5), and the number of superpixel number M. Apart from c and n_s that are fixed as 64 and 3 in our method, Table V summarizes other empirically optimal values for four datasets. Besides, their influence on our SPTNet-NTLA will be studied in Section IV-E.

TABLE V
VALUES OF FREE PARAMETERS FOR DIFFERENT DATASETS

	Indian Pines	University of Pavia	Houston	Houston 2018
$s \times s$	9×9	9×9	9×9	13×13
h	8	8	8	8
τ	0.3	0.5	0.5	0.7
λ_s	0.2	0.9	0.8	0.8
M	800	17000	18000	30000

We select three assessment criteria to evaluate our SPTNet-NTLA numerically [48], i.e., overall accuracy (OA), average accuracy (AA), and kappa coefficient (K). The numerical results displayed in the following are the mean values of five random experiments. Also, the standard deviation is counted for reference. Data augmentation and label smoothing are used in the training phase to ensure stability. In the inference stage, we choose the last rather than the best model to complete the prediction.

C. Classification Performance

To demonstrate the effectiveness of SPTNet-NTLA, we compare it with nine different methods, including SCE [52], normalized cross entropy (NCE) [53], coteaching [63], 3-D CNN [71], RLPA [27], multiscale segmentation-based multilayer spectral–spatial graphs (MSSGs) [57], DCRN [59], superpixel-guided sample selection neural network (S3Net) [58], and TCRL [60]. The first three methods are developed for natural images, while the last six models are HSI-oriented. The natural-scene-oriented methods are chosen, because their effectiveness has been proven in various applications. By applying them to noisy HSI classification tasks, robust comparison benchmarks can be provided to evaluate SPTNet-NTLA's performance comprehensively. Also, the comparisons can be enriched in this case. All compared methods are completed by ourselves using the released codes. The free parameters of different methods are consistent with the original literature, and the experimental settings are the same as the contents mentioned in Sections IV-A and IV-B. To deeply study our SPTNet-NTLA and the compared methods, we count the results when the noise rate r changes from 0.1 to 0.4.

1) Results on Indian Pines: The results of different methods counted on the Indian Pines dataset at different noisy rates are summarized in Table VI. By observing them, we can easily find the following points.

1) First, the performance of all methods is acceptable when r is small. When r increases, their behavior is weakened to a certain degree. Especially, for SCE, NCE, coteaching, and 3-D CNN, the performance drops between $r = 0.1$ and $r = 0.4$ are distinct, over approximately 10%. This is because the accumulated errors caused by the noisy samples will reduce the model's discrimination in classification. Once the number of noisy samples exceeds a certain threshold, the generated accumulated errors would destroy the classification models.
2) Second, the performance of 3-D CNN is superior to two machine learning-based algorithms (i.e., RLPA and

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TABLE VI
CLASSIFICATION PERFORMANCE OF DIFFERENT METHODS COUNTED ON THE INDIAN PINES DATASET AT DIFFERENT NOISY RATES

r	Assessment Criteria	SCE [52]	NCE [53]	Coteaching [63]	3-D-CNN [71]	RLPA [27]	MSSG [57]	DCRN [59]	S3Net [58]	TCRL [60]	**SPTNet-NTLA (Our)**
0.1	OA(%)	92.66±1.41	98.07±0.67	97.16±0.18	87.95±1.09	78.90±0.62	80.60±0.60	97.67±1.28	96.30±0.46	98.23±0.33	**98.61±0.11**
	AA(%)	93.90±1.66	98.75±0.49	98.07±0.54	90.53±0.74	83.29±1.78	79.67±1.58	98.81±0.50	97.62±0.45	90.57±3.71	**99.23±0.17**
	K×100	91.51±1.61	97.76±0.77	96.70±0.20	86.12±1.21	75.68±0.65	77.63±0.66	97.29±1.50	95.70±0.53	97.95±0.38	**98.38±0.11**
0.2	OA(%)	86.04±1.25	97.01±0.74	94.61±1.00	80.58±2.36	78.38±0.47	80.44±0.33	96.89±1.21	96.14±0.36	96.97±0.90	**98.22±0.14**
	AA(%)	86.83±1.81	98.10±0.22	96.28±0.48	82.72±1.70	82.66±1.69	79.69±1.53	98.00±0.84	96.21±2.27	86.01±1.43	**98.94±0.09**
	K×100	83.90±1.40	96.53±0.86	93.75±1.15	77.83±2.54	75.11±0.57	77.45±0.38	96.38±1.41	95.52±0.42	96.48±1.04	**97.93±0.17**
0.3	OA(%)	76.74±1.07	94.30±1.63	94.57±0.60	71.45±1.71	77.87±0.77	79.87±0.67	96.43±1.11	94.98±1.15	95.95±0.94	**97.38±0.21**
	AA(%)	76.62±2.15	94.76±1.38	96.47±0.35	71.81±2.27	81.08±1.56	79.14±1.56	97.94±0.71	96.47±0.57	82.73±0.86	**98.30±0.12**
	K×100	73.44±1.18	93.39±1.86	93.70±0.69	67.61±1.82	74.55±0.76	76.80±0.76	95.84±1.30	94.17±1.32	95.29±1.09	**96.95±0.25**
0.4	OA(%)	67.55±1.31	85.46±2.73	88.37±0.75	63.38±1.63	76.32±1.16	79.07±1.16	93.15±3.45	92.92±1.00	93.85±0.57	**96.46±0.36**
	AA(%)	67.08±1.53	85.33±3.12	92.88±0.63	64.25±1.18	78.56±2.17	78.95±1.40	95.87±1.94	94.89±1.85	81.57±0.54	**97.99±0.18**
	K×100	63.26±1.40	83.30±3.07	86.60±0.84	58.72±1.69	72.87±1.23	75.93±1.28	92.03±4.03	91.82±1.15	92.88±0.66	**95.89±0.42**

TABLE VII
CLASSIFICATION PERFORMANCE OF DIFFERENT METHODS COUNTED ON THE UNIVERSITY OF PAVIA DATASET AT DIFFERENT NOISY RATES

r	Assessment Criteria	SCE [52]	NCE [53]	Coteaching [63]	3-D-CNN [71]	RLPA [27]	MSSG [57]	DCRN [59]	S3Net [58]	TCRL [60]	**SPTNet-NTLA (Our)**
0.1	OA(%)	96.62±0.13	99.69±0.13	98.67±0.31	94.69±0.44	90.96±0.65	90.67±0.49	99.65±0.40	99.65±0.06	99.72±0.16	**99.85±0.01**
	AA(%)	95.60±0.33	99.71±0.10	98.07±0.43	94.06±0.13	92.06±0.31	91.68±0.39	99.48±0.42	99.24±0.12	99.64±0.18	**99.67±0.07**
	K×100	95.39±0.17	99.58±0.18	98.19±0.42	92.80±0.58	87.83±0.84	87.33±0.64	99.52±0.56	99.52±0.08	99.62±0.22	**99.80±0.01**
0.2	OA(%)	90.64±0.68	99.33±0.08	96.84±0.62	90.61±0.77	90.90±0.62	90.62±0.33	99.49±0.51	99.53±0.12	99.64±0.21	**99.79±0.03**
	AA(%)	89.69±0.38	99.28±0.10	96.73±0.65	89.72±0.88	91.94±0.29	91.54±0.28	99.58±0.15	98.89±0.12	99.59±0.18	**99.61±0.06**
	K×100	87.42±0.89	99.08±0.10	95.70±0.83	87.36±1.00	87.37±0.79	87.39±0.42	99.30±0.69	99.35±0.17	99.50±0.28	**99.72±0.05**
0.3	OA(%)	82.48±1.51	98.02±0.63	96.43±0.44	84.06±0.90	90.84±0.26	90.49±0.27	99.16±0.60	99.45±0.11	99.51±0.33	**99.60±0.15**
	AA(%)	81.01±1.03	98.43±0.17	95.30±0.43	82.65±0.59	91.89±0.24	91.25±0.33	99.27±0.17	98.96±0.12	99.50±0.35	**99.52±0.02**
	K×100	76.83±1.87	97.30±0.84	95.14±0.60	78.86±1.14	87.81±0.33	87.22±0.35	98.85±0.82	99.25±0.14	99.33±0.45	**99.46±0.21**
0.4	OA(%)	71.50±0.86	95.19±1.10	94.23±0.38	78.76±2.01	90.35±0.42	90.12±0.29	98.98±0.60	98.90±0.32	99.27±0.21	**99.59±0.08**
	AA(%)	69.69±1.05	95.33±1.34	92.94±0.42	75.36±1.38	91.28±0.25	90.67±0.50	99.16±0.21	98.30±0.51	99.17±0.43	**99.47±0.14**
	K×100	63.21±1.02	93.49±1.49	92.14±0.52	71.89±2.48	87.03±0.54	86.73±0.37	98.62±0.81	98.50±0.44	99.00±0.28	**99.44±0.12**

MSSG) due to the better feature extraction capacity of the deep model when the noise is less. However, since they distinguish noisy samples and accomplish classification in a two-stage manner, RLPA and MSSG outperform, as the noise rate rises. However, as the independent combination of the noisy sample selection scheme and the machine-learning-based classifier cannot achieve satisfactory results, RLPA and MSSG's behavior is weaker than three general-purpose methods, i.e., SCE, NCE, and coteaching. Nevertheless, as they do not consider the specific characteristics of HSIs, their performance is inferior to DCRN, S3Net, and TCRL. Especially, in the case of a significant noisy rate, their advantages are more pronounced.

3) Third, although DCRN, S3Net, and TCRL are stronger than other models, our SPTNet-NTLA outperforms them in all cases. When $r = 0.1$, the improvements achieved by SPTNet-NTLA in OA/AA/K are 0.94%/0.42%/1.09% (over DCRN), 2.31%/1.61%/2.68% (over S3Net), and 0.38%/8.66%/0.43% (over TCRL). When $r = 0.4$, the enhancements in OA/AA/K are 3.31%/2.12%/3.86% (over DCRN), 3.54%/3.10%/4.07% (over S3Net), and 2.61%/16.42%/3.01% (over TCRL). The reasons for these encouraging results are twofold. On the one hand, the proposed SPTNet explores the spectral information within HSIs from local and global aspects. Besides, the redundant knowledge arising from patch inputting is also considered in SPTNet. On the other hand, the developed noise-tolerant learning algorithm not only whittles down the impact of noisy samples on SPTNet but also effectively helps SPTNet mine the rich information hidden in all training data.

The above discussion illustrates the usefulness of SPTNet-NTLA.

2) Results on University of Pavia: The classification performance of different methods on the University of Pavia dataset is exhibited in Table VII. Similar to the investigation of the Indian Pines dataset, the relatively weak models are PLPA, MSSG, and 3-D CNN. Three natural-image-oriented models perform better. However, their shortcomings are noticeable, i.e., their behavior drops dramatically when r rises. The performance of three HSI-oriented networks (DCRN, S3Net, and TCRL) is superior to other compared methods. Our SPTNet-NTLA is also the strongest one among all methods. For instance, when $r = 0.1$, the enhancements of SPTNet-NTLA in K value are 4.41% (over SCE), 0.22% (over NCE), 1.61% (over coteaching), 7.00% (over 3-D CNN), 11.97% (over RLPA), 12.47% (over MSSG), 0.28% (over DCRN), 0.28% (over S3Net), and 0.18% (over TCRL). Besides, the performance gaps (assessed by K value) between SPTNet-NTLA and others are enlarged when $r = 0.4$, i.e., 36.23% (over SCE), 5.95% (over NCE), 7.30% (over coteaching), 27.55% (over 3-D CNN), 12.41% (over RLPA), 12.71% (over MSSG), 0.82% (over DCRN), 0.94% (over S3Net), and 0.44% (over TCRL). The above analysis reflects that our method performs stably from another perspective. Regardless of whether the noise rate is high or low, the performance of our SPTNet-NTLA is positive.

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TABLE VIII
CLASSIFICATION PERFORMANCE OF DIFFERENT METHODS COUNTED ON THE HOUSTON DATASET AT DIFFERENT NOISY RATES

r	Assessment Criteria	SCE [52]	NCE [53]	Coteaching [63]	3-D-CNN [71]	RLPA [27]	MSSG [57]	DCRN [59]	S3Net [58]	TCRL [60]	**SPTNet-NTLA (Our)**
0.1	OA(%)	93.32±0.48	97.94±0.15	97.06±0.44	91.49±1.15	89.95±1.34	90.17±0.94	97.01±0.83	95.92±0.63	98.33±0.35	**98.42±0.14**
	AA(%)	93.73±0.48	98.27±0.08	97.50±0.35	92.04±1.16	90.53±1.17	90.80±0.83	97.08±1.16	96.52±0.49	98.38±0.30	**98.61±0.20**
	K×100	92.77±0.51	97.76±0.16	96.81±0.47	90.78±1.24	88.93±1.46	89.34±1.02	96.76±0.90	95.57±0.69	98.19±0.38	**98.29±0.15**
0.2	OA(%)	84.75±1.30	96.20±0.34	93.70±0.69	86.01±2.06	89.79±1.45	90.02±0.94	96.51±1.25	94.56±0.67	97.94±0.20	**98.00±0.15**
	AA(%)	85.18±1.28	96.97±0.26	94.69±0.49	86.32±2.25	90.55±1.35	90.84±0.77	96.99±0.97	95.43±0.54	98.07±0.46	**98.19±0.11**
	K×100	83.50±1.40	95.88±0.37	93.17±0.75	84.85±2.23	89.10±1.57	89.18±1.02	96.22±1.35	94.10±0.73	97.77±0.22	**97.84±0.16**
0.3	OA(%)	76.23±0.99	93.50±0.79	92.24±0.74	82.71±3.82	89.19±1.09	89.06±0.81	95.75±1.16	92.67±0.99	96.66±0.15	**96.79±0.16**
	AA(%)	77.09±1.07	94.81±0.66	93.42±0.55	82.81±4.30	89.72±1.06	89.84±0.79	96.41±0.92	93.99±0.76	95.45±0.38	**97.31±0.13**
	K×100	74.31±1.07	92.96±0.85	91.60±0.80	81.27±4.13	88.28±1.18	88.14±0.88	95.39±1.26	92.05±1.08	96.38±0.17	**96.52±0.17**
0.4	OA(%)	66.34±1.85	90.92±0.89	91.64±0.98	78.28±2.17	87.58±0.60	87.70±0.79	93.94±0.58	92.33±0.92	96.20±0.97	**96.27±0.09**
	AA(%)	66.30±1.41	91.58±0.67	92.22±0.65	79.54±2.12	88.20±1.00	88.88±0.73	94.60±0.43	93.64±0.70	95.91±1.25	**96.42±0.11**
	K×100	63.65±1.99	90.16±0.96	90.94±1.05	76.50±2.34	86.53±0.65	86.67±0.85	93.43±0.63	91.68±0.99	95.88±1.06	**95.96±0.09**

TABLE IX
CLASSIFICATION PERFORMANCE OF DIFFERENT METHODS COUNTED ON THE HOUSTON 2018 DATASET AT DIFFERENT NOISY RATES

r	Assessment Criteria	SCE [52]	NCE [53]	Coteaching [63]	3-D-CNN [71]	RLPA [27]	MSSG [57]	DCRN [59]	S3Net [58]	TCRL [60]	**SPTNet-NTLA (Our)**
0.1	OA(%)	71.45±0.46	69.42±1.09	65.87±1.58	59.37±1.16	45.66±3.33	47.46±2.81	76.86±1.04	73.14±0.97	77.61±1.72	**81.28±0.48**
	AA(%)	84.29±0.46	85.67±0.40	82.55±0.63	78.00±0.67	70.06±0.58	70.45±1.03	87.76±0.56	86.89±0.44	60.18±1.86	**88.95±0.24**
	K×100	65.27±0.46	63.17±1.17	59.33±1.60	52.38±1.19	39.29±2.85	40.78±2.49	71.34±1.16	67.23±1.06	72.31±1.91	**76.35±0.57**
0.2	OA(%)	61.43±3.12	63.09±1.73	61.43±1.76	55.09±0.90	44.02±2.45	46.98±5.62	75.67±1.00	70.53±0.83	73.52±2.71	**79.49±0.96**
	AA(%)	76.45±0.68	83.25±0.60	80.70±0.78	75.15±0.36	70.21±0.73	69.94±1.27	87.62±0.65	85.69±0.49	55.42±2.19	**88.10±0.37**
	K×100	54.52±2.95	56.51±1.69	54.47±1.56	47.61±0.92	37.98±2.22	40.42±5.05	70.06±1.10	64.39±0.97	67.77±2.89	**74.22±1.13**
0.3	OA(%)	52.43±1.54	58.61±3.01	59.30±0.94	50.49±1.91	42.85±0.67	45.31±2.97	73.70±1.08	66.74±2.26	72.77±2.49	**78.62±0.44**
	AA(%)	67.54±1.08	79.91±0.52	78.53±0.71	73.13±0.98	68.49±1.03	68.09±0.63	85.89±0.34	84.46±0.21	54.53±2.04	**87.60±0.21**
	K×100	45.04±1.30	51.65±2.87	52.06±0.97	43.87±1.74	36.88±0.84	38.71±2.69	67.53±1.20	60.41±2.24	66.58±2.61	**73.20±0.49**
0.4	OA(%)	41.63±2.09	51.45±3.04	49.82±1.22	49.75±1.83	37.33±2.87	42.23±5.62	66.89±4.16	59.65±3.37	65.67±3.65	**76.39±1.20**
	AA	57.64±1.01	73.86±2.18	76.02±0.55	70.79±0.87	65.82±0.42	66.60±0.35	85.39±0.61	81.84±0.97	50.40±2.78	**85.62±0.20**
	K×100	34.51±1.56	44.11±2.82	43.46±0.94	42.74±1.55	32.05±2.30	36.10±4.57	60.55±4.24	53.29±3.11	59.30±3.66	**70.51±1.30**

3) Results on Houston: Table VIII lists the evaluation results for various methods counted on the Houston dataset at different noise levels. Similar to the conclusions obtained on the Indian Pines and University of Pavia datasets, the proposed SPTNet-NTLA is the best model among all approaches. Taking OA values as an example, when r varies from 0.1 to 0.4, the improvements achieved by our method are 5.10%/13.25%/20.56%/29.93% (over SCE), 0.48%/1.80%/3.29%/5.35% (over NCE), 1.36%/4.30%/4.55%/4.63% (over coteaching), 6.93%/11.99%/14.08%/17.99% (over 3-D CNN), 8.47%/8.21%/7.60%/8.69% (over RLPA), 8.25%/7.98%/7.73%/8.57% (over MSSG), 1.41%/1.49%/1.04%/2.33% (over DCRN), 2.50%/3.44%/4.12%/3.94% (over S3Net), and 0.09%/0.06%/0.13%/0.07% (over TCRL). Also, the standard deviations of our SPTNet-NTLA are the smallest in most cases compared with other models. These encouraging results demonstrate that SPTNet-NTLA is helpful in distinguishing the diverse land covers within HSI under the noisy scenario.

4) Results on Houston 2018: The results of various methods counted on Houston 2018 with different noise rates are shown in Table IX. Similar to the conclusions obtained on the other three datasets, although the accuracy of different methods decreases as the noise rate increases, the proposed SPTNet-NTLA performs best. For example, when $r = 0.2$, compared with the three better compared models (i.e., DCRN, S3Net, and TCRL), the improvements obtained by SPTNet-NTLA in OA/AA/K are 3.82%/0.48%/4.16% (over DCRN), 8.96%/2.41%/9.83% (over S3Net), and 5.97%/32.68%/6.45% (over TCRL). When $r = 0.4$, the enhancements in OA/AA/K are as high as 9.50%/0.23%/9.96% (over DCRN), 16.74%/3.78%/17.22% (over S3Net), and 10.72%/35.22%/11.21% (over TCRL). These positive results further illustrate the usefulness of our SPTNet-NTLA.

Apart from the numerical results discussed above, we also display and study the classification maps obtained by different models under four noise rates. The details can be found in the Supplementary Material.

D. Ablation Analysis

The proposed SPTNet-NTLA involves two parts, including a feature extraction backbone (i.e., SPTNet) and the corresponding training method (i.e., the noise-tolerant learning algorithm). This section will inspect them deeply and separately. For clarity, the experiments in this section are conducted under a noise ratio of 0.4 unless otherwise specified.

1) Effectiveness of SPTNet: To study SPTNet's usefulness, we select five popular HSI classification networks and compare the performance of SPTNet with these networks in the noise-free scenario. The chosen networks are 3-D CNN [71], SSRN [72], SpectralFormer [20], SPRLT [24], and SSFTT [21]. Here, 3-D CNN and SSRN are CNN-based models, and SpectralFormer, SPRLT, and SSFTT are transformer-based approaches. Their free parameters are kept consistent with the original paper. The experimental settings are aligned with the contents discussed in Section IV-B. Since there is no noisy sample, the CE loss and the usual learning manner are adopted to train different networks. Table X summarizes their performance on four HSI datasets. The results

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TABLE X
CLASSIFICATION PERFORMANCE OF SPTNET AND OTHER FIVE HSI CLASSIFICATION NETWORKS IN THE NOISE-FREE CONDITION, COUNTED ON FOUR HSI DATASETS

Datasets	Assessment Criteria	3-D-CNN [71]	SSRN [72]	SpectralFormer [20]	SPRLT [24]	SSFTT [21]	SPTNet
Indian Pines	OA(%)	96.19±0.29	97.84±0.72	96.08±0.49	96.88±0.64	97.18±0.25	**98.36±0.16**
	AA(%)	97.76±0.30	98.64±0.24	96.10±1.01	97.62±0.61	98.25±0.28	**99.20±0.17**
	K×100	95.58±0.34	97.48±0.84	95.45±0.56	96.37±0.74	96.73±0.29	**98.09±0.19**
University of Pavia	OA(%)	99.57±0.07	99.69±0.19	99.05±0.11	99.50±0.13	99.57±0.08	**99.86±0.04**
	AA(%)	99.38±0.10	99.71±0.08	98.63±0.14	99.45±0.06	98.96±0.33	**99.80±0.08**
	K×100	99.41±0.10	99.58±0.26	98.70±0.15	99.32±0.18	99.42±0.10	**99.81±0.06**
Houston	OA(%)	96.90±0.53	96.73±1.88	96.72±0.65	97.90±0.24	99.23±0.17	**99.27±0.19**
	AA(%)	97.33±0.40	97.42±1.45	97.17±0.45	98.32±0.21	99.39±0.13	**99.43±0.15**
	K×100	96.64±0.57	96.46±2.04	96.44±0.70	97.72±0.26	99.16±0.19	**99.21±0.21**
Houston 2018	OA(%)	72.21±0.71	77.15±1.50	64.67±0.89	75.18±2.04	78.75±1.58	**81.87±0.62**
	AA(%)	86.84±0.39	87.38±1.35	78.84±0.43	86.10±0.57	88.27±0.18	**89.02±0.09**
	K×100	66.25±0.82	71.60±1.67	57.23±0.91	69.15±2.34	73.45±1.72	**77.07±0.73**

show that SPTNet outperforms the other five networks in all cases. Taking the Indian Pines dataset as an example, the improvements achieved by SPTNet in OA values are 2.17% (over 3-D CNN), 0.52% (over SSRN), 2.28% (over SpectralFormer), 1.48% (over SPRLT), and 1.18% (over SSFTT). The positive results indicate that our SPTNet is competitive in HSI classification.

2) Importance of SPT: The core component of SPTNet is SPT. To explore its utility, we replace it with the standard transformer encoder, swin transformer [73], pyramid vision transformer [74], and SPRLT [24] and construct SPTNet's variants. The new deep models are recorded as TNet, STNet, PVTNet, and SPRLTNet for short. Their behavior on the noisy HSI classification is summarized in Fig. 7. All networks are trained by the proposed noise-tolerant learning algorithm, and the experimental settings are the same as the contents mentioned in Section IV-B. We can find that SPTNet achieves the best performance by observing these bars, which implies our SPT is superior to the other transformer encoders. Compared with the other four transformer encoders focusing on mining global context within HSI, our SPT also reduces the confused clues caused by HSI patches. Therefore, the final noisy classification performance can be enhanced. Those encouraging results demonstrate the advantages of SPT.

3) Significance of the Noise-Tolerant Learning Algorithm: To investigate if our noise-tolerant learning algorithm works or not, we apply it and the normal learning manner (with the CE loss) to train SPTNet and the other five HSI classification networks (mentioned in Section IV-D1) separately under the noisy scenario. Their classification results counted on four datasets are exhibited in Table XI. Note that the free parameters of the selected networks are set according to the original literature, and the experimental settings maintain consistency with the contents mentioned in Section IV-B.

From the observation of the results, we can find the following conclusion. First, compared with the normal learning manner, the proposed noise-tolerant algorithm helps different networks achieve stronger behavior when noisy samples exist. Besides, the performance gaps between the same model with different learning algorithms are distinct. These discoveries

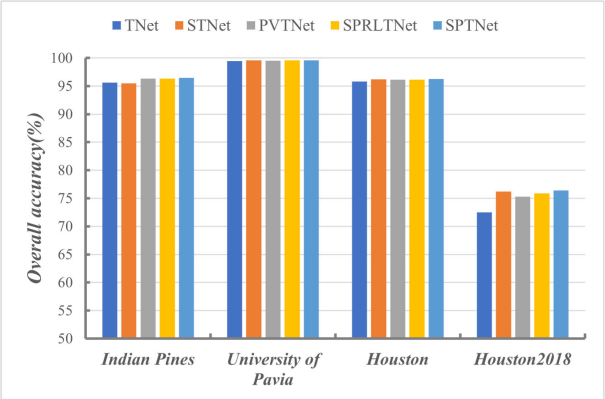


Fig. 7. Classification performance of SPTNet and its four variants counted on four HSI datasets when noise rate equals 0.4.

denote the usefulness of the proposed noise-tolerant learning algorithm. Second, when we use the usual learning method, the performance of SPTNet is relatively weak. Nevertheless, when the proposed learning algorithm is applied, SPTNet performs best among all networks. The reasons behind this are as follows. When we use the usual learning manner, the behavior of SPTNet is stronger than that of the other five networks in the noise-free condition (see Section IV-D1). This indicates that SPTNet can extract more information from HSIs than other networks. Because of this characteristic, SPTNet is more susceptible to obtaining erroneous clues when noisy samples are present. As a result, its performance experiences a noticeable decline. Once using our noise-tolerant learning algorithm, the performance improvement of SPTNet is the most significant, which implies the superiorities of our learning algorithm and the potential of SPTNet in noisy classification simultaneously.

4) Superiority of the Noise-Tolerant Learning Algorithm: After discussing the significance of the proposed learning algorithm, we will study its advantages in this section. To this end, four noise-resistant algorithms are chosen to train SPTNet, including SCE loss (SCELoss) [52], noise-tolerant deep neighbor embedding (NTDEN) [75], NCE loss and RCE loss

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TABLE XI

CLASSIFICATION PERFORMANCE OF VARIOUS NETWORKS BASED ON THE NORMAL TRAINING MANNER AND OUR NOISE-TOLERANT ALGORITHM. THE RESULTS ARE COUNTED ON FOUR HSI DATASETS UNDER THE NOISY RATE OF 0.4

Datasets		Indian Pines		University of Pavia		Houston		Houston 2018	
Learning Manner		Normal	Noise-Tolerant	Normal	Noise-Tolerant	Normal	Noise-Tolerant	Normal	Noise-Tolerant
3-D-CNN [71]	OA(%)	63.38±1.63	86.72±0.71	78.76±2.01	98.95±0.20	78.28±2.17	90.58±0.52	49.40±1.82	54.07±1.43
	AA(%)	64.25±1.18	86.50±0.54	75.36±1.38	98.75±0.25	79.54±2.12	92.17±0.46	74.22±1.26	78.48±0.22
	K×100	58.72±1.69	84.71±0.82	71.89±2.48	98.57±0.28	76.50±2.34	89.79±0.57	42.83±1.74	47.15±1.35
SSRN [72]	OA(%)	84.36±2.30	94.05±0.92	84.84±5.68	98.63±0.61	87.86±1.39	94.33±0.84	58.71±3.50	64.79±2.30
	AA(%)	89.56±1.42	93.47±2.89	89.47±3.71	98.00±1.27	89.05±1.16	95.06±0.67	77.18±2.47	81.45±0.32
	K×100	82.12±2.49	93.11±1.06	80.28±7.18	98.13±0.83	86.86±1.50	93.85±0.90	51.50±3.56	57.03±2.29
SpectralFormer [20]	OA(%)	82.04±1.05	87.03±0.28	87.61±1.89	97.70±0.20	85.12±0.23	90.88±0.98	44.42±2.24	50.97±1.80
	AA(%)	83.21±2.00	87.24±0.31	91.23±0.94	96.65±0.17	86.15±0.20	92.15±0.75	66.56±0.70	72.47±0.96
	K×100	79.40±1.16	85.17±0.32	83.71±2.35	96.86±0.27	83.88±0.25	90.12±1.06	36.73±1.85	43.18±1.76
SPRLT [24]	OA(%)	82.71±3.18	93.51±0.71	92.99±1.84	99.31±0.13	92.21±0.81	92.71±0.51	52.60±1.442	53.06±2.31
	AA(%)	88.01±2.72	90.87±2.43	96.33±0.49	99.22±0.09	93.49±0.61	94.02±0.42	76.68±0.98	78.38±0.50
	K×100	80.28±3.47	92.48±0.82	90.65±2.41	99.06±0.18	91.55±0.88	92.10±0.55	45.62±1.36	45.91±2.05
SSFTT [21]	OA(%)	78.39±2.31	95.09±0.43	77.89±1.77	98.33±0.48	85.58±2.38	96.16±0.17	45.61±2.56	68.76±0.93
	AA(%)	85.16±0.74	96.60±0.52	83.18±0.70	98.01±0.32	87.92±1.80	96.40±0.20	64.04±0.39	83.17±0.64
	K×100	75.50±2.52	94.31±0.49	71.51±2.00	97.72±0.66	84.40±2.57	95.83±0.18	38.13±2.20	61.97±0.94
SPTNet	OA(%)	76.07±0.97	**96.46±0.36**	77.50±2.86	**99.59±0.08**	82.10±1.41	**96.27±0.09**	52.32±2.96	**76.39±1.20**
	AA(%)	84.35±0.54	**97.99±0.18**	84.42±1.67	**99.47±0.14**	84.02±1.21	**96.42±0.11**	64.18±1.24	**85.62±0.20**
	K×100	72.87±1.07	**95.89±0.42**	71.27±3.42	**99.44±0.12**	80.64±1.52	**95.96±0.09**	44.09±2.67	**70.51±1.30**

(NRLoss) [59], and sample selection and sample correction (SSSC) [58]. The trained SPTNets' performance for noisy HSI classification is summarized in Table XII. The results of SPTNet trained by our learning algorithm are also exhibited for reference. It is not difficult to find that SPTNet's behavior is the strongest when our learning algorithm is used. This is because our algorithm combines the cross-selection strategy and similarity loss, which facilitates the discovery of clean samples and fully uses the entire sample to guarantee the classification result.

5) Contributions of Components in the Noise-Tolerant Learning Algorithm: Our algorithm encompasses three components, i.e., the data partition scheme, cotraining mechanism, and similarity regularization. We individually add them to a "Base" training method to study their contributions. Here, "Base" means we use training data to train SPTNet with the CE loss without considering the noisy samples. The results of SPTNets trained by different algorithms are summarized in Table XIII. Not surprisingly, SPTNet trained by "Base" performs worst due to the adverse impacts of noisy samples. When we add the data partition scheme, SPTNet's behavior becomes stronger, which implies the importance of the data partition scheme. After including the cotraining mechanism, the performance of SPTNet is enhanced to a certain degree. This indicates that the selected clean samples are more credible under the cotraining paradigm. Once the similarity regularization is added, SPTNet performs best on different HSIs. This is owing to the similarity regularization that helps SPTNet explore profitable information from both clean and noisy samples without considering the manual labels. The above ablation studies demonstrate that all components of the noise-tolerant learning algorithm play an irreplaceable role of their own.

E. Impacts of Free Parameters

As mentioned in Section IV-B, there are seven free parameters, including the patch size $s \times s$, the volume of attention heads h in the transformer encoder, the dimension of the features c in SFE, the number of SPT n_s, the hyperparameter λ_s for the final loss (11), the threshold τ in the data partition scheme (5), and the number of superpixels M. In this section, we will analyze their influence by changing their values. Note that only one parameter will be tuned each time, and other parameters are set according to Table V. For clarity, we fix $r = 0.4$ and select OA as the assessment criterion.

1) Analysis of Patch Size: To study the impact of patch size $s \times s$ on our method, we change its values at a range of $[7 \times 7, 15 \times 15]$. The performance of different SPTNet-NTLAs with various $s \times s$ counted on four datasets is shown in Figs. 8(a), 9(a), 10(a), and 11(a). By observing them, it is easy for us to find that the behavior of SPTNet-NTLAs becomes stronger first and then shifts weaker. For the first three datasets, their peaks occur at 9×9. For Houston 2018, its peak appears at 13×13. This is because of the following: 1) the contents within small patches are not rich enough to train our model and 2) large patches contain more redundant messages that would confuse the proposed model. Here, large and small are relative concepts defined based on image size. Therefore, we suggest that the suitable sizes of HSI patches are 13×13 (Houston 2018) and 9×9 (others).

2) Study of the Number of Attention Heads: In this section, we set h equal to 2, 4, 8, and 16, respectively, to observe its impact. The results counted on four datasets are depicted in Figs. 8(b), 9(b), 10(b), and 11(b). The OA values of SPTNet-NTLA become larger when h increases from 2 to 8. Then, they slightly decrease when h changes from 8 to 16. Fortunately, the SPTNet-NTLA's behavior does not vary significantly. In other words, SPTNet-NTLA is not sensitive to h. Considering the trade-off between the performance and time costs, we suggest $h = 8$.

3) Study of the Dimension of the Features in the SFE: As mentioned in Section III-B, we fix the dimension of outputs of SFE as 64. In this section, we change this value from 16 to 128 to study its impact on classification. The results of

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TABLE XII

CLASSIFICATION PERFORMANCE OF SPTNET COMBINED WITH DIFFERENT NOISE-RESISTANT ALGORITHMS AND OUR NOISE-TOLERANT ALGORITHM ON FOUR HSI DATASETS AT A NOISE RATE OF 0.4

Datasets	Assessment Criteria	SCELoss [52]	NTDEN [75]	NRLoss [59]	SSSC [58]	SPTNet-NTLA (Our)
Indian Pines	OA(%)	93.55±0.93	93.64±0.69	94.02±0.21	94.19±0.48	**96.46±0.36**
	AA(%)	90.08±0.12	96.46±0.50	94.18±1.14	94.87±2.83	**97.99±0.18**
	K×100	92.53±1.07	92.64±0.80	92.81±0.23	93.27±0.56	**95.89±0.42**
University of Pavia	OA(%)	98.65±0.34	98.68±0.12	98.45±0.50	98.53±0.68	**99.59±0.08**
	AA(%)	98.61±0.34	98.89±0.17	98.76±0.28	98.70±0.20	**99.47±0.14**
	K×100	98.16±0.46	98.19±0.16	97.89±0.68	98.00±0.92	**99.44±0.12**
Houston	OA(%)	93.47±0.91	94.30±1.35	93.41±1.03	95.82±0.30	**96.27±0.09**
	AA(%)	94.38±0.69	95.04±1.21	94.01±0.60	96.14±0.66	**96.42±0.11**
	K×100	92.92±0.98	93.82±1.46	92.85±1.12	95.47±0.68	**95.96±0.09**
Houston 2018	OA(%)	66.82±1.04	63.59±1.18	66.24±3.40	68.73±1.32	**76.39±1.20**
	AA(%)	84.65±0.34	81.40±1.04	80.83±1.72	82.42±0.44	**85.62±0.20**
	K×100	60.02±1.02	56.54±1.18	59.24±3.51	61.90±1.41	**70.51±1.30**

TABLE XIII

RESULTS OF ABLATION STUDIES OF VARIOUS PARTS OF OUR NOISE-TOLERANT ALGORITHM ON FOUR DATASETS

Datasets	Base	Data Partition	Co-training	Similarity Regularization	OA(%)	AA(%)	K×100
Indian Pines	✓				76.07±0.97	84.35±0.54	72.87±1.07
	✓	✓			95.58±0.86	97.51±0.26	94.87±0.99
	✓	✓	✓		95.76±0.80	97.72±0.33	95.08±0.92
	✓	✓	✓	✓	**96.46±0.36**	**97.99±0.18**	**95.89±0.42**
University of Pavia	✓				77.50±2.86	84.42±1.67	71.27±3.42
	✓	✓			99.32±0.19	99.25±0.13	99.07±0.26
	✓	✓	✓		99.46±0.23	99.32±0.24	99.26±0.32
	✓	✓	✓	✓	**99.59±0.08**	**99.47±0.14**	**99.44±0.12**
Houston	✓				82.10±1.41	84.02±1.21	80.64±1.52
	✓	✓			94.05±0.82	94.86±0.89	93.54±0.89
	✓	✓	✓		95.54±0.58	96.20±0.56	95.17±0.63
	✓	✓	✓	✓	**96.27±0.09**	**96.42±0.11**	**95.96±0.09**
Houston 2018	✓				52.32±2.96	64.18±1.24	44.09±2.67
	✓	✓			73.65±1.29	83.88±0.54	67.35±1.41
	✓	✓	✓		76.10±1.57	85.55±0.28	70.20±1.70
	✓	✓	✓	✓	**76.39±1.20**	**85.62±0.20**	**70.51±1.30**

SPTNets with different SFE outputs counted on four datasets are shown in Figs. 8(c), 9(c), 10(c), and 11(c). By observation, we can see that the trend of curves is to increase first and then decrease. Also, the pinnacle values appear when the dimension of SFE outputs is equal to 64 in all cases. The reasons behind this can be summarized as follows. When the feature dimension is small, the learned features cannot express complex contents within HSIs. However, when the feature dimension is large, the obtained features contain much more useless clues that harm the downstream classification. Therefore, we suggest that the output feature dimension of SFE is 64.

4) Study of the Number of SPT: To investigate the impact of the number of SPT n_s on SPTNet performance, we vary the number of SPT from 1 to 4 and observe the behavior of SPTNet. The results counted on four datasets are shown in Figs. 8(d), 9(d), 10(d), and 11(d). By observing those curves, we can find that the OA values first increase and then decrease as the number of SPT increases. Also, the peak values appear when the number of SPT equals 3. This is because when there are fewer SPTs, the model's ability to capture spatial information is insufficient, resulting in less satisfactory classification results. However, when there are too many SPTs, the model is prone to overfitting, which, in turn, limits its classification effectiveness. Taking all factors into consideration, we recommend using three SPTs.

5) Impact of the Hyperparameter: To analyze the influence of λ_s on our model, we vary its value from 0.1 to 0.9 with an interval of 0.1. The OA values of different SPTNet-NTLAs are summarized in Figs. 8(e), 9(e), 10(e), and 11(e). For the Indian Pines dataset, SPTNet-NTLA's performance improves sharply when λ_s varies from 0.1 to 0.2, remains when λ_s increases

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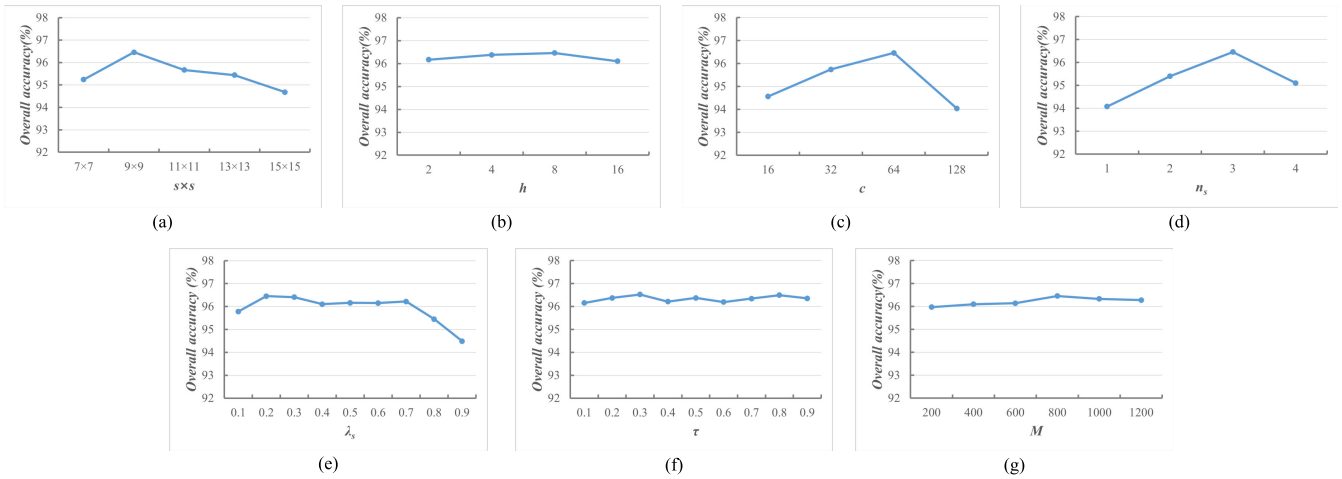


Fig. 8. Influence of different free parameters on our SPTNet-NTLA. The OA curves are counted on the Indian Pines dataset. (a) $s \times s$. (b) h. (c) c. (d) n_s. (e) λ_s. (f) τ. (g) M.

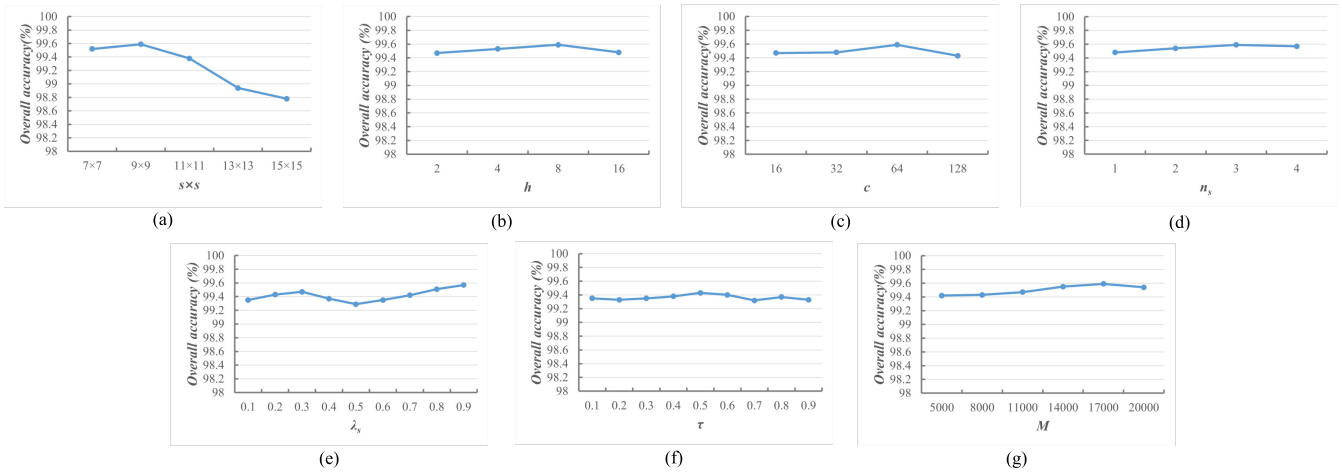


Fig. 9. Influence of different free parameters on our SPTNet-NTLA. The OA curves are counted on the University of Pavia dataset. (a) $s \times s$. (b) h. (c) c. (d) n_s. (e) λ_s. (f) τ. (g) M.

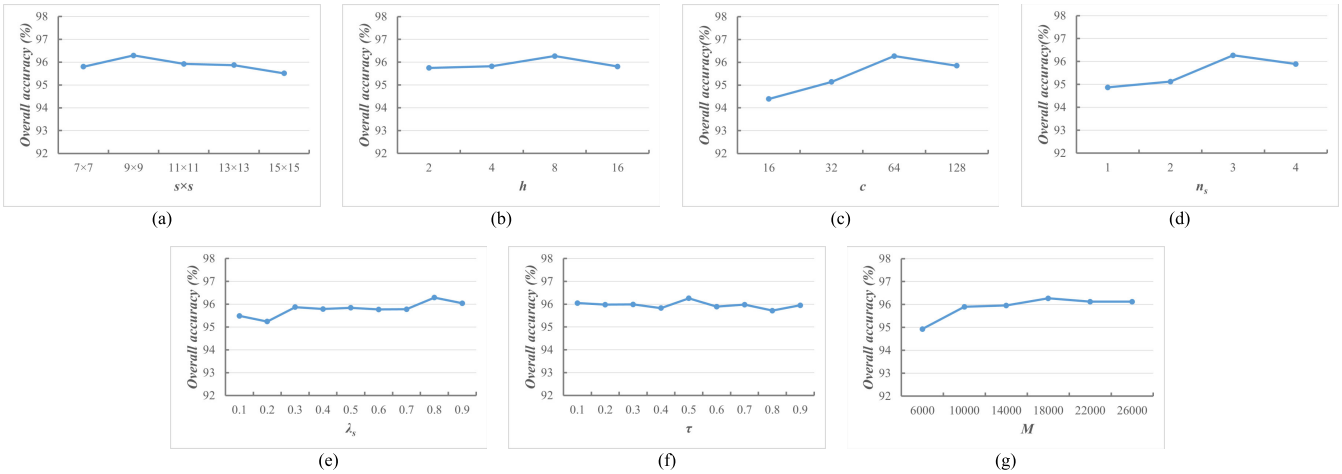


Fig. 10. Influence of different free parameters on our SPTNet-NTLA. The OA curves are counted on the Houston dataset. (a) $s \times s$. (b) h. (c) c. (d) n_s. (e) λ_s. (f) τ. (g) M.

from 0.2 to 0.7, and drops when $\lambda_s \geq 0.7$. Besides, the peak OA value appears at $\lambda_s = 0.2$. The above investigation means that CE loss dominates the final classification, and the similarity loss provides important assistance. For the University of Pavia dataset, the SPTNet-NTLA's performance shows an upward trend with λ_s rises, and the best SPTNet-NTLA appears at $\lambda_s = 0.9$. This means the function of the similarity loss is significant. A similar conclusion can

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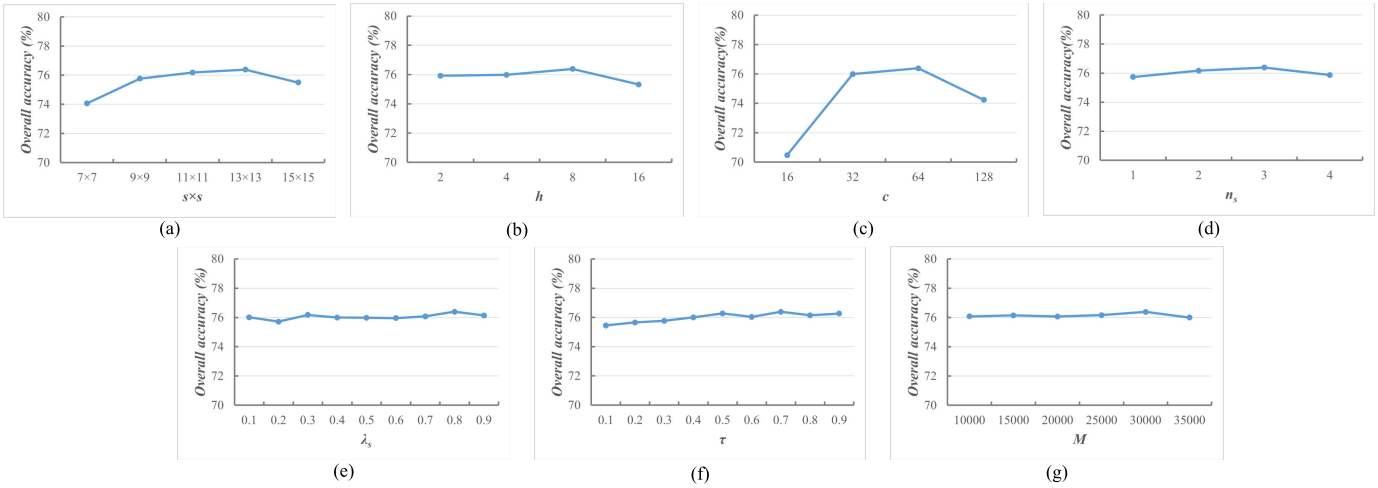


Fig. 11. Influence of different free parameters on our SPTNet-NTLA. The OA curves are counted on the Houston 2018 dataset. (a) $s \times s$. (b) h. (c) c. (d) n_s. (e) λ_s. (f) τ. (g) M.

be obtained for the Houston and Houston 2018 datasets, i.e., larger λ_s better results. Besides, the highest OA of these two HSIs emerges at $\lambda_s = 0.8$. In summary, the above observations denote that two terms in the final loss (11) are complementary and indispensable to each other.

6) Effect of the Threshold: This parameter determines the volume of the clean samples. A smaller τ means more clean samples would be used to calculate the CE loss (6). To study its effect, we tune it in a range of $[0.1, 0.9]$. The OA curves of SPTNet-NTLAs with different τ values counted on four datasets are shown in Figs. 8(f), 9(f), 10(f), and 11(f). For the Indian Pines dataset, the OA values of SPTNet-NTLAs fluctuate between 96% and 97%, and the best value appears at $\tau = 0.3$. For the University of Pavia and Houston datasets, the OA values of various SPTNet-NTLAs are concentrated around 99.4% and 96%, respectively. Also, the best values for these two datasets appear at $\tau = 0.5$. For the Houston 2018 dataset, the best OA value of SPTNet-NTLAs occurs at $\tau = 0.7$. The above observations imply that the impacts of τ on SPTNet-NTLA are small for the University of Pavia and Houston datasets but are relatively big for the Indian Pines and Houston 2018 datasets. This is mainly due to the characteristics of the data itself. Compared with the other three datasets, the size of the Indian Pines dataset is relatively small, as well as the scale of the training set. In addition to the class imbalance problem, SPTNet requires more labeled samples to learn meaningful information. Consequently, the optimal τ is small, so more labeled samples can be selected as the clean data. The Houston 2018 dataset is the largest among the four HSIs, and its categories are the richest. In this case, the noisy labels' influence would be significant. Therefore, it is reasonable for SPTNet-NTLA to use a more strict selection strategy (i.e., a larger τ value). For the other two datasets, since the scale of labeled samples is large, the class imbalance issue is not prominent, and the categories are not too complex, the most proper τ emerges at 0.5.

7) Influence of the Numbers of Superpixels: The number of superpixels M is crucial for the similarity regularization proposed in Section III-C. When M is small, the obtained homogeneous regions will contain too many pixels, which harms establishing similar pixel pairs. When M is large, the volume of pixels within homogeneous regions would be small. The constructed similar pixel pairs cannot provide enough clues for mining the similarity relations. Considering the differences in sizes among different datasets, we change M from 200 to 1200 for the Indian Pines dataset, from 5000 to 20 000 for the University of Pavia dataset, from 6000 to 26 000 for the Houston dataset, and from 10 000 to 35 000 for the Houston 2018 dataset. In this way, each superpixel will contain approximately 10–40 pixels. The different SPTNet-NTLAs' OA values counted on four datasets are exhibited in Figs. 8(g), 9(g), 10(g), and 11(g). From the observation of these OA curves, we can find that their changes are relatively gradual, and the optimal number of superpixels for the four datasets is 800, 17 000, 18 000, and 30 000, respectively. The above inspection implies that the proper volume of pixels in a homogeneous region is around 20–30.

F. Complexity Analysis

In this section, we study the efficiency of the proposed SPTNet-NTLA from the following three points. First, the model size of SPTNet is counted and exhibited in Table XIV. The deep-learning-based compared models' sizes are also counted for reference. By comparison, we can observe that the proposed SPTNet is close to the lightest, whose size is as small as 0.68 MB. This is attributed to the powerful information-learning ability of SPT, which ensures that we do not need more layers to capture messages hidden in HSIs. Besides, introducing spatial pooling further reduces the SPTNet's size by removing redundant information. Second, the training times of different methods counted on the Indian Pines dataset are summarized in Table XIV. The most time-consuming model is S3Net, and the most time-saving method is DCRN. The training time of our SPTNet-NTLA is slightly longer than most of the models. The main reason is that we adopt the cotraining idea and train two SPTNets simultaneously. Fortunately, once the SPTNets are trained by the noise-tolerant learning

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TABLE XIV
MODEL SIZE, TRAINING TIME, AND INFERENCE TIME OF EACH ALGORITHM ON THE INDIAN PINES DATASET

Methods	Model Size (MB)	Training Time (s)	Inference Time (s)
SCE [52]	6.70	388.27	8.71
NCE [53]	6.70	442.62	8.31
Coteaching [63]	17.80	536.23	8.11
3-D-CNN [71]	23.15	2830.36	46.31
DCRN [59]	2.84	**307.41**	11.21
S3Net [58]	23.15	5698.53	47.26
TCRL [60]	**0.62**	854.41	1.58
SPTNet-NTLA (Our)	0.68	902.97	**1.51**

algorithm, SPTNet-NTLA's inference time is the shortest (see the last column of Table XIV). The above contents illustrate that our SPTNet-NTLA is a light and efficient noisy HSI classification method.

V. CONCLUSION

This article proposes a simple yet effective network SPTNet and a noise-tolerant learning algorithm for the HSI classification task with noisy labels. SPTNet first analyzes the spectral information within HSIs by combining 2-D CNN and 3-D CNN. Then, a new SPT is developed based on the standard transformer to extract global spatial contexts hidden in HSIs and remove confusing information caused by patch input. After weaving the above techniques into an end-to-end framework, the complex contents of HSIs can be thoroughly mined. For the noise-tolerant learning algorithm, a data partition scheme is designed to divide the training data into clean and noisy sets. Also, the cotraining mechanism is adopted to ensure the credibility of selected samples. Then, the clean samples are used to train SPTNets with classification loss. Meanwhile, considering that the knowledge of noisy samples is also helpful for understanding HSIs' land covers, we develop a similarity regularization to mine the messages covered by noisy samples. Finally, SPTNets can be trained under the noisy scenario by jointly optimizing the classification and similarity losses. Encouraging experimental results counted on four public HSI datasets validate the effectiveness of our proposed methods on the noisy HSI classification.

In real scenarios, noisy samples' distributions are more complex, which may be feature-dependent or category-dependent. Our future work is to conquer these challenges and ensure classification performance.

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